

A hybrid PSO-GA algorithm for constrained optimization problems

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ABSTRACT

The main objective of this paper is to present a hybrid technique named as a PSO-GA for solving the constrained optimization problems. In this algorithm, particle swarm optimization (PSO) operates in the direction of improving the vector while the genetic algorithm (GA) has been used for modifying the decision vectors using genetic operators. The balance between the exploration and exploitation abilities have been further improved by incorporating the genetic operators, namely, crossover and mutation in PSO algorithm. The constraints defined in the problem are handled with the help of the parameter-free penalty function. The experimental results of constrained optimization problems are reported and compared with the typical approaches exist in the literature. As shown, the solutions obtained by the proposed approach are superior to those of existing best solutions reported in the literature. Furthermore, experimental results indicate that the proposed approach may yield better solutions to engineering problems than those obtained by using current algorithms.

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1. Introduction

Constraint optimization forms an important part of any problem in engineering and industry. Most of the real world optimization problems have constraints of different types which modify the shape of search space. During the last decades, a wide variety of meta-heuristic algorithm have been designed and applied to solve the constrained optimization problems. As constrained optimization problems are difficult to solve than unconstrained optimization problems due to the presence of various types of constraints (in the form of equalities or inequalities) and their interrelationship between the objective functions. These types of problems have normally mixed, continuous or discrete, nonlinear objective functions and nonlinear constraints. For optimizing such types of problems or functions, there are two general methods, namely, mathematical programming and meta-heuristic methods. Various mathematical programming methods such as linear programming, homogenous linear programming, integer programming, dynamic programming, and nonlinear programming have been applied for solving such type of problems. These methods use gradient information to search the solution space near an initial starting point. In general, gradient-based methods converge faster and can obtain solutions with higher accuracy as compared to stochastic approaches during fulfilling the local search task. However, for effective implementation of these methods, the variables and cost function of the generators need to be continuous. Furthermore, a good starting point is vital for these methods to be executed successfully. In many optimization problems, prohibited zones, side limits, and non-smooth or non-convex cost functions need to be considered. As a result, these

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non-convex optimization problems cannot be solved by the traditional mathematical programming methods. Although dynamic programming or mixed integer non-linear programming and their modifications offer few options in solving non-convex problems but they require considerable high computational cost.

As an alternative to the conventional mathematical approaches, the meta-heuristic optimization techniques have been used to obtain global or near-global optimum solutions. Due to their capability of exploring and detecting promising regions in the search space at an accurate time, these methods are quite suitable for global searches. Furthermore, these methods lighten the need for continuous cost functions and variables used for mathematical optimization methods. Although these are approximate methods, i.e., their solution is good, but not necessarily optimal, they do not require the derivatives of the objective function and/or constraints and employ probabilistic transition rules instead of deterministic ones. Therefore, the researchers focus on the heuristic techniques that seek the good constructive solution within a reasonable time [1]. However, the traditional or heuristic algorithms require derivative for all the non-linear constraint functions to evaluate the system performance. But in real-life problem, it is difficult to derive due to high computational complexity of the system. Thus to overcome this, nature inspired algorithms are the latest state of art algorithms. These algorithms work well with optimization problems as well as other problems than the classical methods because classical methods are inflexible in nature. It has been proved by many researchers that nature inspired algorithms namely, genetic algorithms (GAs), particle swarm optimization (PSO), Harmony search (HS) and evolution strategies (ES) are attractive because they do not apply mathematical assumptions to the optimization problems and have better global search abilities over conventional optimization algorithms [2–9]. These meta-heuristic techniques have been applied in various fields of the constrained optimization problems, namely, engineering problems, structural design problems, reliability optimization, decision making etc. For instance, GA has been combined with other evolutionary algorithms to attain more efficiency from a computational point of view. Coello [10], Dimopoulos [11] applied genetic algorithms to solve these mixed-integer engineering design optimization problems. Coello and Montes [12] proposed a dominance-based selection scheme to incorporate constraints into the fitness function of a genetic algorithm used for global optimization. Tsai [13] proposed a novel method to solve non-linear fractional programming problems occurring in engineering design and management. Montes et al. [14] presented a modified version of the differential evolution algorithm to solve design problems in which a criteria based on feasibility and a diversity mechanism are used to maintain infeasible solution. Omran and Salman [2] presented a new parameter-free meta-heuristic algorithm, named as CODEQ, that is a hybrid of concepts from chaotic search, opposition-based learning, differential evolution and quantum mechanics. He et al. [3], He and Wang [4], Shi and Eberhart [15] applied particle swarm optimization to solve these mixed-integer design optimization problems. Moreover, Coelho [16] presented a quantum-behaved PSO (QPSO) approaches using mutation operator with Gaussian probability distribution while Cagnina et al. [7] introduced a simple constraint particle swarm optimization (SiC-PSO) algorithm to solve constrained engineering optimization problems. Lee and Geem [17], Fesanghary et al. [18] applied harmony search algorithm for these types of problems. Kaveh and Talatahari [5] developed a hybridized algorithm based on the particle swarm optimization with the passive congregation (PSOPC), the ant colony algorithm (ACO), and the harmony search (HS) approach, so-called HPSACO to solve the design optimization problems. Gandomi et al. [19,20] used a cuckoo search and Firefly algorithm to solve mixed continuous/discrete structural optimization problems. Mehta and Dasgupta [21] presented a methodology for unconstrained optimization problem in which constraints are handled with the Nelder and Mead Simplex search method. Kaveh and Talatahari [22] presented an improved ant colony optimization for solving the constrained engineering design problems consisting of continuous or discrete search domains. Hedar and Fukushima [23] proposed an efficient simulated-annealing-based method called filter simulated annealing for solving the constraint global optimization problem. In their approach, filter-set-based procedure is built in their proposed approach and finally an intensification scheme is applied as a final stage in order to overcome the slow convergence of simulated-annealing-based methods. Garg [24] solved the structured engineering design problems using penalty guided artificial bee colony algorithm.

As demonstrated in the literature, the aforementioned optimization techniques are successfully applied to various constrained optimization problems. However, the performance while getting an optimal or near to optimal solution shows a significant difference. Therefore, the previously developed algorithms, as stochastic optimization techniques, heuristic algorithm have some weakness such as the lower robustness, premature convergence of the solution, not using a prior knowledge, not exploiting local search information and difficulty in dealing with large scale optimization problems. To overcome this difficulty, meta-heuristics/ evolutionary algorithms have been selected and successfully applied to handle a number of optimization problems [19,24]. The vast majority of heuristic and meta-heuristic algorithms have been derived from the behavior of biological systems and/or physical systems in nature. The two main tasks for any meta-heuristic algorithms are searching around the current best solutions and then select the best candidates from them. Secondly, it makes sure that the algorithm can explore the search space more efficiently rather than by randomization. Based on these important tasks, the meta-heuristic algorithms are preferable methodologies than the others for solving specific optimization problems. Modern meta-heuristic algorithms have been developed with an aim to carry out a global search with three main purposes: solving problems faster, solving large problems, and obtaining robust algorithms. GA and PSO are typical examples of these algorithms and have been successfully applied to solve the various engineering design optimization problem efficiently. However, both the algorithms, GA and PSO, have strengths and weaknesses. In GA, if an individual is not selected then the information contained by that individual is lost, but PSOs have memory. However, without a selection operator, PSOs may waste resources on poor individuals. While GAs has trouble finding an exact solution and are good at reaching a global region, a PSO's group interactions boosts the search for an optimal solution. Thus keeping the advantages of both algorithms into account, the main objective of this paper is to develop an efficient hybrid technique named PSO-GA to find the optimal solution of the constraint nonlinear optimization problems.

Therefore, in this study the applicability of evolutionary and/or socially inspired algorithms to multimodal, mixed-variable constrained optimization problems, is investigated. For this, a hybridized technique named as a PSO-GA has been presented by taking the advantages of both GA and PSO algorithm for solving the nonlinear design optimization problems. In this approach, PSO operates in the direction of improving the vector while GA has been used for modifying the decision vectors using genetic operators. The proposed algorithms have been tested with some well-known mathematical and engineering optimization problems. Numerical results reveal that the proposed algorithm is a powerful search algorithm for various optimization problems. The remaining content of this paper is organized as follows. [Section 2](#) presents the entire notations that have been used in the manuscript. [Section 3](#) deals with the general formulation of the optimization problem and method for handling the constraints. [Section 4](#) elaborates the proposed hybrid PSO-GA approach as well as a brief overview of GA and PSO algorithms. The computational results of the structural design problem are discussed and compared with existing methods in [Section 5](#). Finally, the concrete conclusions drawn are discussed in [Section 6](#).

2. Notations

The following notations are used in the entire paper

D	number of decision variables.
M	number of constraints.
S	feasible search space.
f_w	worst feasible solution.
g_j	j th constraint function.
$f(\cdot)$	cost function.
γ	decreasing rate of no. of individuals that effected by GA.
β	increasing rate of GA maximum iteration.
GA_{PS}	population size of the particles in GA.
$GA_{MinIter}$	minimum number of iteration in GA.
$GA_{MaxIter}$	maximum number of iteration in GA.
GA_{Num}	current number of individuals that effected by GA.
GA_{NumMin}	minimum number of individuals that effected by GA.
GA_{NumMax}	maximum number of individuals that effected by GA.
GA_{MinPS}	first population size in GA
GA_{MaxPS}	last population size in GA
PSO_i	current PSO iteration.
$PSO_{MaxIter}$	maximum number of iteration in PSO.

3. Non-linear constrained optimization problem

A general non-linear constrained optimization problem is defined as follows.

$$\begin{aligned}
 &\text{Minimize } f(x) \\
 &\text{subject to } h_k(x) = 0 \quad ; \quad k = 1, 2, \dots, p \\
 &\quad \quad \quad g_j(x) \leq 0 \quad ; \quad j = 1, 2, \dots, q \\
 &\quad \quad \quad l_i \leq x_i \leq u_i \quad ; \quad i = 1, 2, \dots, n
 \end{aligned} \tag{1}$$

where $x = [x_1, x_2, \dots, x_n]^T$ denotes the n -dimensional vector of decision variables; f is the objective function; l_i and u_i are the minimum and maximum permissible values for the i th variable respectively; p is the number of equality constraints and q is the number of inequality constraints. Let $S = \{x \in \mathbb{R} \mid g_j(x) \leq 0; j = 1, 2, \dots, p + q (= M)\}$ be the set of the feasible solutions, so the problem (1) can be formulated as:

$$\begin{aligned}
 &\text{Minimize } f(x) \\
 &\text{subject to } g_j(x) \leq 0 \quad ; \quad j = 1, 2, \dots, M \\
 &\quad \quad \quad l_i \leq x_i \leq u_i \quad ; \quad i = 1, 2, \dots, n
 \end{aligned} \tag{2}$$

The main task while solving the constrained optimization problem is to handle the constraints. In the constrained optimization problem, it is not an easy to find the feasible solution of the problem due to the presence of both types of constraints in the form of an equalities and inequalities. To handle these constraints, many different approaches have been proposed. The most common approach in the evolutionary algorithm community is to make use of penalty functions. When using a penalty function, the amount of constraint violation is used to punish or penalize an infeasible solution so that feasible solutions are favored by the selection process. Despite the popularity of penalty functions, they have several drawbacks out of which the main one is that of having too many parameters to be adjusted and finding the right combination of the same may not be easy. Also during that, the search is very slow and there is no guarantee that the optima will be attained. To overcome this limitation, Deb [25] modified these algorithms using the concept of parameter free penalty functions, i.e. one attempts to solve an unconstrained problem in a

Algorithm 1 Pseudo code of Genetic algorithm (GA).

```

1: Objective function:  $f(\mathbf{x})$ 
2: Define Fitness  $F$  (eg.  $F \propto f(\mathbf{x})$  for maximization)
3: Initialize population
4: Initial probabilities of crossover ( $p_c$ ) and mutation ( $p_m$ )
5: do
6:   Generate new solution by crossover and mutation
7:   if  $p_c > \text{rand}$ , Crossover; end if
8:   if  $p_m > \text{rand}$ , Mutate; end if
9:   Accept the new solution if its fitness increases.
10:  Select the current best for the next generation.
11: While maximum iterations or minimum error criteria is not attained

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search space S using a modified objective function F such as

$$F(\mathbf{x}) = \begin{cases} f(\mathbf{x}) & \text{if } \mathbf{x} \in S \\ f_w + \sum_{j=1}^M g_j & \text{if } \mathbf{x} \notin S \end{cases} \quad (3)$$

where $\mathbf{x}$ is the set of solutions obtained by approaches and f_w is the worst feasible solution in the population.

4. Hybrid PSO-GA approach

This section describes the novel hybridized genetic algorithm and particle swarm optimization for solving the constrained optimization problems.

4.1. Genetic algorithm

Genetic Algorithms (GAs) are adaptive heuristic search algorithms introduced in the evolutionary themes of natural selection. The fundamental concept of the GA design is to model processes in a natural system that is required for evolution, specifically those that follow the principles posed by Charles Darwin to find the survival of the fittest [1]. GAs constitutes an intelligent development of the random search within a defined search space to solve a problem. GAs were first pioneered by John Holland in the 1960s, and has been widely studied, experimented, and applied in numerous engineering disciplines. GA was introduced as a computational analogy of adaptive systems. They are modeled loosely on the principles of the evolution through natural selection, employing a population of individuals that undergo selection in the presence of variability-inducing operators such as mutation and recombination (crossover). A fitness function is used to evaluate individuals, and reproductive success varies with fitness. The pseudo code of the GA algorithm is described in Algorithm 1:

4.2. Particle swarm optimization

In 1995, Eberhart and Kennedy [26,27] developed PSO, a population based on stochastic optimization strategy, inspired by social behavior of a flock of birds, schools of fish, a swarm of bees and even sometimes social behavior of human. Though PSO is similar to Genetic Algorithm (GA) in terms of population initialization with random solutions and searching for global optima in successive generations, PSO does not undergo crossover and mutation, whereas the particles move through the problem space following the current optimum particles. The underlying concept is that, for every time instant, the velocity of each particle also known as the potential solution, changes between its personal best (pbest) and global best (gbest) locations. Mathematically, swarm of particles is initialized randomly over the search space and move through D -dimensional space to search for new solutions. Let x_k^i and v_k^i respectively be the position and velocity of i th particle in the search space at k th iteration, then its velocity and position of this particle at $(k+1)$ th iteration are updated using the following equations:

$$v_{k+1}^i = \underbrace{w \cdot v_k^i}_{\text{inertia}} + \underbrace{c_1 \cdot r_1 \cdot (p_k^i - x_k^i)}_{\text{personal influence}} + \underbrace{c_2 \cdot r_2 \cdot (p_k^g - x_k^i)}_{\text{social influence}} \quad (4)$$

$$x_{k+1}^i = x_k^i + v_{k+1}^i \quad (5)$$

where r_1 and r_2 represent random numbers between 0 and 1, c_1 and c_2 are constants, p_k^i represents the best ever position of i th particle, and p_k^g corresponds to the global best position in the swarm up to k th iteration.

The essential steps of the particle swarm optimization can be summarized as the pseudo code given in Algorithm 2.

Algorithm 2 Pseudo code of Particle swarm optimization (PSO).

- 1: Objective function: $f(x)$, $x = (x_1, x_2, \dots, x_D)$;
- 2: Initialize particle position and velocity for each particle and set $k = 1$.
- 3: Initialize the particle's best known position to its initial position i.e. $p_k^i = x_k^i$.
- 4: **do**
- 5: Update the best known position (p_k^i) of each particle and swarm's best known position (p_k^g).
- 6: Calculate particle velocity according to the velocity equation (4).
- 7: Update particle position according to the position equation (5).
- 8: **While maximum iterations or minimum error criteria is not attained**

4.3. Hybrid PSO-GA approach

The motivation of developing PSO-GA approach is to combine the advantage of the genetic algorithm and particle swarm optimization. By embedding of the genetic operators in the standard PSO, the balance between the exploration and exploitation ability is improved further. However, both the models have strengths and weaknesses. In GA, if an individual is not selected the information contained by that individual is lost, but PSOs have memory. However, without a selection operator, PSOs may waste resources on poor individuals. Thus, the basic idea in PSO-GA is to combine the ability of social thinking in PSO with the local search capability of GA. As GA and PSO both are population-based algorithm, the hybrid PSO-GA approach is also a population-based algorithm and hence proceed to find the global solution. The approach has been started from the initialization phase, where the particles of the swarm and their corresponding velocities are generated randomly over the search space, i.e. initial position x_0^i of the i th particles randomly taken $x_0^i \sim U(x_{\min}, x_{\max})$ from a uniform distribution in the range $[x_{\min}, x_{\max}]$, where $x_{\min}$ and $x_{\max}$ represent the design variables' lower and upper bounds respectively. However, the velocity vector is used to update the current position of each particle in the swarm and it is based on the "memory" gained by the each particle, as well as the knowledge gained by the swarm as a whole. Thus, the position of each particle in the swarm is adjusted according to its own experience and that of its neighbors. The former one is known as personal best (pbest) position and the latter one is gbest position. After the iteration, position of each particle is updated according to equation (5). After forming a new generation in PSO iterations, some particles of new population are selected and then GA is applied for each of them separately. Since, population size of the particle is very large, so in order to save the time, GA is not applied to the whole population. So, the question arises, how many swarms should be evolved in a PSO renewal generation? For this, out of total population size of the swarm, the number is defined by GA_{Num} in equation (6) are taken which evolved in each PSO generation.

$$GA_{Num} = GA_{NumMax} - \left(\frac{PSO_i}{PSO_{MaxIter}} \right)^{\gamma} \times (GA_{NumMax} - GA_{NumMin}) \quad (6)$$

where PSO_i represents the current PSO iteration and $PSO_{MaxIter}$ be the maximum number of generations in PSO. After selecting the best individual from the population, the algorithm aims to create a new population by replacing points in the current population with better points via genetic principles i.e. by undergoing selection, crossover and mutation operators. One point crossover has been employed for recombination of the two parents using roulette wheel selection. After the selection, crossover and mutation operations, a form of elitism is performed to maintain the best solutions in the population with the following equation:

$$x_{i+1} = \begin{cases} y_i & \text{if } f(y_i) < f(x_i) \\ x_i & \text{otherwise} \end{cases} \quad i = 1, 2, \dots, GA_{PS} \quad (7)$$

After the evaluation of the new population, population size and maximum iteration numbers for GA changes with respect to iteration of PSO and their relationship is defined as follow:

$$GA_{PS} = GA_{MinPS} + \left(\frac{PSO_i}{PSO_{MaxIter}} \right)^{\gamma} \times (GA_{MaxPS} - GA_{MinPS}) \quad (8)$$

$$GA_{MaxIter} = GA_{MinIter} + \left(\frac{PSO_i}{PSO_{MaxIter}} \right)^{\beta} \times (GA_{MaxIter} - GA_{MinIter}) \quad (9)$$

Through a repeated process of reproduction the population is guided towards the global optimum. The complete representation of the proposed method is presented in Fig. 1.

5. Numerical examples

In order to validate the proposed algorithm, several examples have been taken from the optimization literature to demonstrate how the proposed approach works. These examples have linear and non-linear constraints, and have been previously solved by using a variety of other algorithms, which are useful to determine the quality of the solutions produced by the proposed approach. The presented algorithm is implemented in Matlab (MathWorks) and the program has been run on a T6400 @

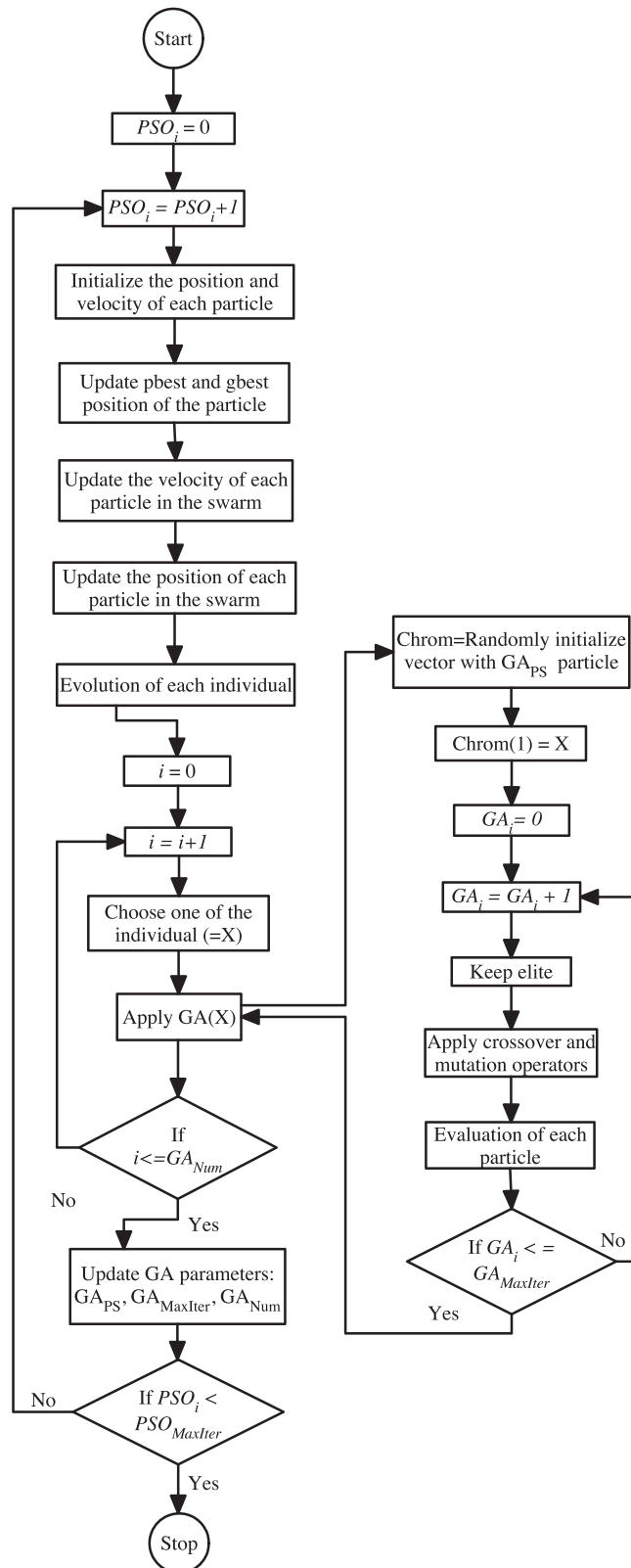


Fig. 1. Hybrid PSO-GA algorithm.

Table 1

Optimal results for Himmelblau's nonlinear optimization problem (NA means not available).

Version	Methods	Design variables						Constraints		
		x_1	x_2	x_3	x_4	x_5	$f(X)$	$0 \leq g_1 \leq 92$	$90 \leq g_2 \leq 110$	$20 \leq g_3 \leq 25$
I	Himmelblau [28]	NA	NA	NA	NA	NA	−30373.9490	NA	NA	NA
	Homaifar et al. [29]	80.39	35.07	32.05	40.33	33.34	−30005.700	91.65619	99.53690	20.02553
	Deb [25]	NA	NA	NA	NA	NA	−30665.539	NA	NA	NA
	Lee and Geem [17]	78.00	33.00	29.995	45.00	36.776	−30665.500	92.00004 ^a	98.84051	19.99994 ^a
	He et al. [3]	78.00	33.00	29.995256	45.00	36.7758129	−30665.539	93.28536 ^a	100.40478	20.00000
	Dimopoulos [11]	78.00	33.00	29.995256	45.00	36.775813	−30665.54	92.00000	98.84050	20.00000
	Gandomi et al. [19]	78.00	33.00	29.99616	45.00	36.77605	−30665.233	91.99996	98.84067	20.0003
	Mehta and Dasgupta [21]	78.00	33.00	29.995256	45.00	36.775813	−30665.538741	NA	NA	NA
	Present study	78.00	33.00	29.9951741	45.00	36.7757340	−30665.56614	91.99999	98.84047	20.000
	Shi and Eberhart [15]	78.00	33.00	27.07099	45.00	44.969	−31025.561	93.28533 ^a	100.40473	19.99997 ^a
II	Coello [38]	78.5958	33.01	27.6460	45.00	45.0000	−30810.359	91.956402	100.54511	20.251919
	Coello [10]	78.0495	33.007	27.081	45.00	44.94	−31020.859	93.28381 ^a	100.40786	20.00191
	Hu et al. [30]	78.00	33.00	27.070997	45.00	44.9692425	−31025.5614	92	100.404784	20
	Fesanghary et al. [18]	78.00	33.00	27.085149	45.00	44.925329	−31024.3166	93.27834 ^a	100.39612	20.00000
	Omran and Salman [2]	78.00	33.00	27.0709971	45.00	44.9692425	−31025.55626	93.28536 ^a	100.40478	20.00000
	Present study	78.00	33.00	27.0709505	45.00	44.9691668	−31025.57471	91.99999	100.40476	20.000

^a violate constraints

2 GHz Intel Core(TM) 2 Duo processor with 2 GB of Random Access Memory(RAM). In order to eliminate stochastic discrepancy, in each example, 30 independent runs are made which involves 30 different initial trial solutions with a randomly generated population of size $20 \times D$ for the optimization, where D is the dimension of the problem. The termination criterion has been set either limited to a maximum number of generations (500) or to the order of relative error equal to 10^{-6} , whichever is achieved first. The other specific parameters of algorithms are given below:

PSO settings: Cognitive and social components (c_1 and c_2 in (4)) are constants that can be used to change the weighting between personal and population experience, respectively. In our experiments, cognitive and the social components are both set to 1.5. Inertia weight (w) is represented by the linear expression $w = w_{\max} - (w_{\max} - w_{\min})(\text{iter}/\text{iter}_{\max})$. Here, $w_{\max} = 0.9$ and $w_{\min} = 0.4$ respectively the initial and final values of inertia weight, $\text{iter}_{\max}$ represents the maximum generation number and 'iter' is used as a generation number.

GA settings: In our experiments, we employed a real coded standard GA having an evaluation, fitness scaling, crossover, mutation units. Single point crossover operation with the rate of 0.85 is employed. Mutation operation restores genetic diversity lost during the application of reproduction and crossover. Mutation rate is set to be 0.02. The other randomly control parameters for the proposed approach are taken as $\text{GA}_{\text{MinPS}} = 10$; $\text{GA}_{\text{MinIter}} = 10$; $\text{GA}_{\text{NumMax}} = 20$; $\text{GA}_{\text{NumMin}} = 1$; $\gamma = 10$; $\beta = 15$.

5.1. Constrained optimization problem

5.1.1. Himmelblau's non-linear optimization problem

Before solving the structural engineering problems, the proposed approach was benchmarked using a well-known problem, namely, Himmelblau's problem. This problem has originally been proposed by Himmelblau [28] and it has been widely used as a benchmark non-linear constrained optimization problem. In this problem, there are five positive design variables $X = [x_1, x_2, x_3, x_4, x_5]$, six non-linear inequality constraints, and ten boundary conditions. The problem can be stated as follows:

$$\text{Minimize } f(X) = 5.3578547x_3^2 + 0.8356891x_1x_5 + 37.293239x_1 - 40792.141$$

$$\begin{aligned} \text{s.t. } & 0 \leq g_1(X) \leq 92 \\ & 90 \leq g_2(X) \leq 110 \\ & 20 \leq g_3(X) \leq 25 \end{aligned}$$

$$\text{where } g_1(X) = 85.334407 + 0.0056858x_2x_5 + \mathbf{0.0006262}x_1x_4 - 0.0022053x_3x_5$$

$$g_2(X) = 80.51249 + 0.0071317x_2x_5 + 0.0029955x_1x_2 - 0.0021813x_3^2$$

$$g_3(X) = 9.300961 + 0.0047026x_3x_5 + 0.0012547x_1x_3 + 0.0019085x_3x_4$$

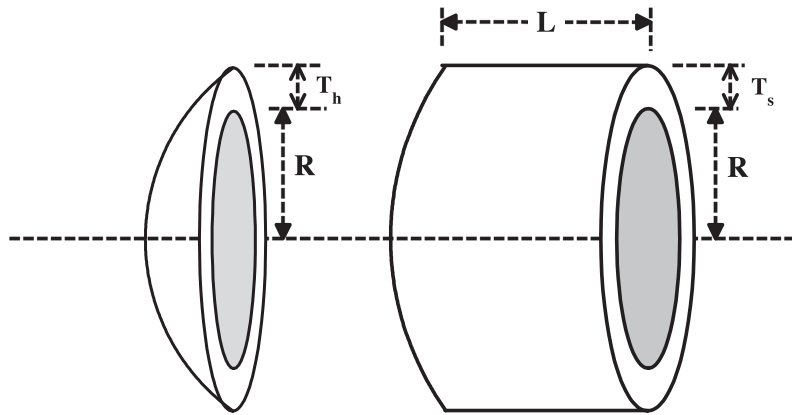
$$78 \leq x_1 \leq 102; \quad 33 \leq x_2 \leq 45; \quad 27 \leq x_3, x_4, x_5 \leq 45$$

This problem was originally proposed by Himmelblau [28], and it has been used before as a benchmark by employing several other methods such as GA [25,29], harmony search [17], PSO [3], Cuckoo search [19], simplex search [21] etc. Also, some of the authors [2,10,15,18,30] have tested their algorithm with another variation of this problem (named as version II), where a parameter 0.0006262 (typeset bold in the constraint g_1) has been taken as 0.00026. The presented algorithm has been tested on both the versions and compares the best solution of the problem with previous best solution reported by them in Table 1. It has been observed from the Table 1 that the solutions given in [2,3,10,15,17,18] are infeasible as they violate the g_1 constraints. The

Table 2

Statistical results for the Himmelblau's problem (NA means not available).

Version	Methods	Best	Median	Mean	Worst	Std.
I	Deb [25]	−30665.537	−30665.535	NA	−29846.654	NA
	Lee and Geem [17]	−30665.500	NA	NA	NA	NA
	He et al. [3]	−30665.539	NA	−30643.989	NA	70.043
	Dimopoulos [11]	−30665.54	NA	NA	NA	NA
	Gandomi et al. [19]	−30665.2327	NA	NA	NA	11.6231
	Mehta and Dasgupta [21]	−30665.538741	NA	NA	NA	NA
	Present study	−30665.566141	−30665.487911	−30665.398373	−30664.624696	0.24084
II	Coello [10]	−31020.859	−31017.21369	−30984.240703	−30792.407737	73.633536
	Hu et al. [30]	−31025.56142	NA	−31025.561420	NA	0
	Fesanghary et al. [18]	−31024.3166	NA	NA	NA	NA
	Omran and Salman [2]	−31025.55626	NA	−31025.556264	NA	NA
	Present study	−31025.574717	−31025.561141	−31025.557816	−31025.492054	0.01526

**Fig. 2.** Pressure vessel design problem figure.

proposed method could obtain the better solution in the variable version I is $X = [78.00, 33.00, 29.99517417, 45.00, 36.7757340]$ with the objective function value -30665.56614 while in version II, the optimal solution is $X = [78.00, 33.00, 27.07095, 45.00, 44.9691668]$ with corresponding function value is -31025.57471 . The best, the average, the worst and the standard deviation of objective function values obtained by 30 runs are reported in Table 2. Based on the above simulation results and comparisons, it can be concluded that proposed PSO-GA approach is robust in nature and has better searching quality. Moreover, the worst solution obtained by the proposed method is still better than the best one obtained by the other methods.

5.2. Structural optimization problems

5.2.1. Design of pressure vessel

A compressed air storage tank with working pressure of 2000 psi and maximum volume of 750ft³. A cylindrical vessel is capped at both ends by hemispherical heads as shown in Fig. 2. Using rolled steel plate, the shell is made in two halves that are joined by two longitudinal welds to form a cylinder. The objective is to minimize the total cost, including the cost of material, forming and welding [31]. There are four design variable associated with it, namely as the thickness of the pressure vessel, $T_s = x_1$, thickness of the head, $T_h = x_2$, inner radius of the vessel, $R = x_3$, and length of the vessel without heads, $L = x_4$ i.e. the variable vectors are given (in inches) by $X = (T_s, T_h, R, L) = (x_1, x_2, x_3, x_4)$. Then, the mathematical model of the problem is summarized as

$$\begin{aligned}
 &\text{Minimize } f(X) = 0.6224x_1x_3x_4 + 1.7781x_2x_3^2 + 3.1661x_1^2x_4 + 19.84x_1^2x_3 \\
 &\text{s.t. } g_1(X) = -x_1 + 0.0193x_3 \leq 0 \\
 &\quad g_2(X) = -x_2 + 0.0095x_3 \leq 0 \\
 &\quad g_3(X) = -\pi x_3^2x_4 - \frac{4}{3}\pi x_3^3 + 1296000 \leq 0 \\
 &\quad g_4(X) = x_4 - 240 \leq 0 \\
 &\quad 1 \times 0.0625 \leq x_1, x_2 \leq 99 \times 0.0625, 10 \leq x_3, x_4 \leq 200
 \end{aligned}$$

The various authors given in [4,5,10,12,16,19,22,31–34] have solved the problem using different algorithms such as genetic adaptive search, a branch and bound method, evolution strategy, a co-evolutionary particle swarm optimization,

Table 3

Comparison of the best solution for pressure vessel design problem found by different methods.

Method	Design variables				Cost $f(X)$
	x_1	x_2	x_3	x_4	
Sandgren [33]	1.125000	0.625000	47.700000	117.701000	8129.1036
Kannan and Kramer [31]	1.125000	0.625000	58.291000	43.690000	7198.0428
Deb and Gene [32]	0.937500	0.500000	48.329000	112.67900	6410.3811
Coello [10]	0.812500	0.437500	40.323900	200.000000	6288.7445
Coello and Montes [12]	0.812500	0.437500	42.097398	176.654050	6059.946
He and Wang [4]	0.812500	0.437500	42.091266	176.746500	6061.0777
Montes and Coello [34]	0.812500	0.437500	42.098087	176.640518	6059.7456
Kaveh and Talatahari [5]	0.812500	0.437500	42.103566	176.573220	6059.0925
Kaveh and Talatahari [22]	0.812500	0.437500	42.098353	176.637751	6059.7258
Zhang and Wang [39]	1.125000	0.625000	58.290000	43.6930000	7197.7000
Cagnina et al. [7]	0.812500	0.437500	42.098445	176.6365950	6059.714335
Coello [16]	0.812500	0.437500	42.098400	176.6372000	6059.7208
He et al. [3]	0.812500	0.437500	42.098445	176.6365950	6059.7143
Lee and Geem [17]	1.125000	0.625000	58.278900	43.75490000	7198.433
Montes et al. [14]	0.812500	0.437500	42.098446	176.6360470	6059.701660
Hu et al. [30]	0.812500	0.437500	42.098450	176.6366000	6059.131296
Gandomi et al. [19]	0.812500	0.437500	42.0984456	176.6365958	6059.7143348
Akay and Karaboga [40]	0.812500	0.437500	42.098446	176.636596	6059.714339
Garg [24]	0.7781977	0.3846656	40.3210545	199.9802367	5885.4032828
Present study	0.7781686	0.3846491	40.3196187	200.000000	5885.3327736

etc. On the other hand, the optimal solution obtained by the proposed approach for the considered problem is $X = [0.7781686, 0.3846491, 40.3196187, 200.00000]$ with corresponding function value equal to $f(X) = 5885.33277$ and constraints $[g_1, g_2, g_3, g_4] = [0, 0, -4.656 \times 10^{-10}, -40]$. The comparison of the proposed results with the existing methodology, results are shown in Table 3 while their corresponding statistical simulation results are summarized in Table 4. The results obtained by using proposed algorithm is better optimized than any other earlier solutions reported in the literature. It has been notified from the Table 4 that the worst solution found by proposed algorithm is better than any of the solution produced by any of the other techniques. In addition, the standard deviation of the results in 30 independent runs is very small.

5.2.2. Welded beam design problem

The welded beam structure, shown in Fig. 3 taken from Rao [35], is a practical design problem that has been often used as a benchmark problem. The objective is to find the minimum fabricating cost of the welded beam subject to constraints on shear stress (τ), bending stress in the beam (θ), buckling load on the bar (P_c), the end deflection of the beam (δ), and side constraints. There are four design variables associated with this problem, namely, thickness of the weld $h = x_1$, length of the welded joint $l = x_2$, the width of the beam $t = x_3$ and thickness of the beam $b = x_4$ i.e. the decision vector is $X = (h, l, t, b) = (x_1, x_2, x_3, x_4)$.

The mathematical formulation of the objective functions $f(X)$ which is the total fabricating cost mainly comprised of the set-up, welding labor, and material cost, are as follows:

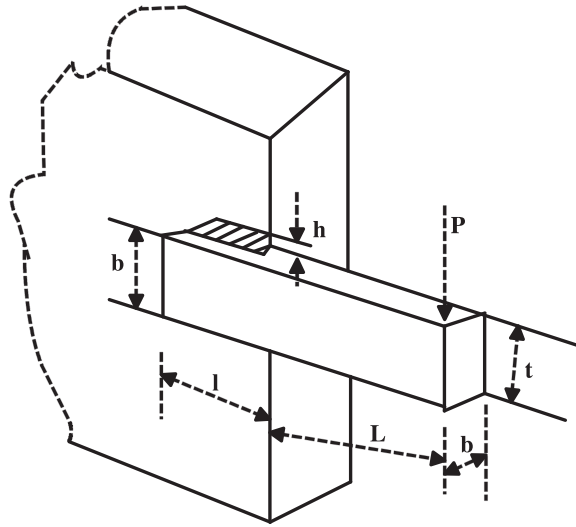
$$\begin{aligned}
 \text{Minimize } f(X) &= 1.10471x_1^2x_2 + 0.04811x_3x_4(14 + x_2) \\
 \text{s.t. } g_1(X) &= \tau(X) - \tau_{\max} \leq 0 \\
 g_2(X) &= \sigma(X) - \sigma_{\max} \leq 0 \\
 g_3(X) &= x_1 - x_4 \leq 0 \\
 g_4(X) &= 0.125 - x_1 \leq 0 \\
 g_5(X) &= \delta(X) - 0.25 \leq 0 \\
 g_6(X) &= P - P_c(X) \leq 0 \\
 g_7(X) &= 0.10471x_1^2 + 0.04811x_3x_4(14 + x_2) - 5 \leq 0 \\
 0.1 \leq x_1 \leq 2; \quad &0.1 \leq x_2 \leq 10; \quad 0.1 \leq x_3 \leq 10; \quad 0.1 \leq x_4 \leq 2
 \end{aligned}$$

where τ is the shear stress in the weld, $\tau_{\max}$ is the allowable shear stress of the weld ($= 13600$ psi), σ the normal stress in the beam, $\sigma_{\max}$ is the allowable normal stress for the beam material ($= 30000$ psi), P_c is the bar buckling load, P the load ($= 6000$ lb), and δ the beam end deflection. The first constraint g_1 ensures that the maximum developed shear stress is less than the allowable shear stress of the weld material. The second constraint, g_2 , checks that the maximum developed normal stress is lower than the allowed normal stress in the beam. The third constraints, g_3 , that the beam thickness ensures exceeds that of the weld. The fourth constraint, g_4 , checks that the weld thickness is above given minimum, and the constraints g_5 , is to ensure that the end deflection of the beam is less than the predefined amount. The sixth constraints, g_6 , make sure that the load on the beam

Table 4

Statistical results of different methods for pressure vessel (NA means not available).

Method	Best	Mean	Worst	Std. Dev.	Median
Sandgren [33]	8129.1036	N/A	N/A	N/A	NA
Kannan and Kramer [31]	7198.0428	N/A	N/A	N/A	NA
Deb and Gene [32]	6410.3811	N/A	N/A	N/A	NA
Coello [10]	6288.7445	6293.8432	6308.1497	7.4133	NA
Coello and Montes [12]	6059.9463	6177.2533	6469.3220	130.9297	NA
He and Wang [4]	6061.0777	6147.1332	6363.8041	86.4545	NA
Montes and Coello [34]	6059.7456	6850.0049	7332.8798	426.0000	NA
Kaveh and Talatahari [22]	6059.7258	6081.7812	6150.1289	67.2418	NA
Kaveh and Talatahari [5]	6059.0925	6075.2567	6135.3336	41.6825	NA
Gandomi et al. [19]	6059.714	6447.7360	6495.3470	502.693	NA
Cagnina et al. [7]	6059.714335	6092.0498	NA	12.1725	NA
Coello [16]	6059.7208	6440.3786	7544.4925	448.4711	6257.5943
He et al. [3]	6059.7143	6289.92881	NA	305.78	NA
Akay and Karaboga [40]	6059.714339	6245.308144	NA	205	NA
Garg [24]	5885.403282	5887.557024	5895.126804	2.745290	5886.14928
Present study	5885.332773	5885.382053	5885.486467	0.049080	5885.3812

**Fig. 3.** Welded beam design problem.

is not greater than the allowed buckling load. The shear stress τ has two components namely primary stress (τ_1) and secondary stress (τ_2) given as

$$\tau(X) = \sqrt{\tau_1^2 + 2\tau_1\tau_2\left(\frac{x_2}{2R}\right) + \tau_2^2} \quad ; \quad \tau_1 = \frac{P}{\sqrt{2}x_1x_2} \quad ; \quad \tau_2 = \frac{MR}{J},$$

where

$$M = P\left(L + \frac{x_2}{2}\right) \quad ; \quad J(X) = 2\left\{\sqrt{2}x_1x_2\left[\frac{x_2^2}{4} + \left(\frac{x_1 + x_3}{2}\right)^2\right]\right\}$$

are known as moments and polar moment of inertia respectively while the other terms associated with the model are as follows

$$R = \sqrt{\frac{x_2^2}{4} + \left(\frac{x_1 + x_3}{2}\right)^2} \quad ; \quad \sigma(X) = \frac{6PL}{x_4x_3^2} \quad ; \quad \delta(X) = \frac{6PL^3}{Ex_3^3x_4}$$

$$P_c(X) = \frac{4.013E\sqrt{\frac{x_3^2x_4^6}{36}}}{L^2}\left(1 - \frac{x_3}{2L}\sqrt{\frac{E}{4G}}\right)$$

$$G = 12 \times 10^6 \text{ psi}, \quad E = 30 \times 10^6 \text{ psi}, \quad P = 6000 \text{ lb}, \quad L = 14 \text{ in}$$

Table 5

Comparison of the best solution for welded beam found by different methods.

Method	Design variables				Cost
	x_1	x_2	x_3	x_4	$f(X)$
Coello [10]	0.208800	3.420500	8.997500	0.210000	1.748309
Coello and Montes [12]	0.205986	3.471328	9.020224	0.206480	1.728226
Hu et al. [30]	0.20573	3.47049	9.03662	0.20573	1.72485084
Hedar and Fukushima [23]	0.205644261	3.472578742	9.03662391	0.2057296	1.7250022
He and Wang [4]	0.202369	3.544214	9.048210	0.205723	1.728024
Dimopoulos [11]	0.2015	3.5620	9.041398	0.205706	1.731186
Mahdavi et al. [36]	0.20573	3.47049	9.03662	0.20573	1.7248
Montes et al. [14]	0.205730	3.470489	9.036624	0.205730	1.724852
Montes and Coello [34]	0.199742	3.612060	9.037500	0.206082	1.73730
Cagnina et al. [7]	0.205729	3.470488	9.036624	0.205729	1.724852
Fesanghary and Mahdavi [18]	0.20572	3.47060	9.03682	0.20572	1.7248
Kaveh and Talatahari [5]	0.205729	3.469875	9.036805	0.205765	1.724849
Kaveh and Talatahari [22]	0.205700	3.471131	9.036683	0.205731	1.724918
Gandomi et al. [20]	0.2015	3.562	9.0414	0.2057	1.73121
Mehta and Dasgupta [21]	0.20572885	3.47050567	9.03662392	0.20572964	1.724855
Akay and Karaboga [40]	0.205730	3.470489	9.036624	0.205730	1.724852
Garg [24]	0.20572450	3.25325369	9.03664438	0.20572999	1.69526388
Present study	0.2057296	3.2531200	9.0366239	0.2057296	1.6952471

Table 6

Statistical results of different methods for welded beam design problem (NA means not available).

Method	Best	Mean	Worst	Std. dev.	Median
Coello [10]	1.748309	1.771973	1.785835	0.011220	NA
Coello and Montes [12]	1.728226	1.792654	1.993408	0.07471	NA
Dimopoulos [11]	1.731186	NA	NA	NA	NA
He and Wang [4]	1.728024	1.748831	1.782143	0.012926	NA
Hedar and Fukushima [23]	1.7250022	1.7564428	1.8843960	0.0424175	NA
Montes et al. [14]	1.724852	1.725	NA	1E-15	NA
Montes and Coello [34]	1.737300	1.813290	1.994651	0.070500	NA
Cagnina et al. [7]	1.724852	2.0574	NA	0.2154	NA
Kaveh and Talatahari [22]	1.724918	1.729752	1.775961	0.009200	NA
Kaveh and Talatahari [5]	1.724849	1.727564	1.759522	0.008254	NA
Gandomi et al. [20]	1.7312065	1.8786560	2.3455793	0.2677989	NA
Mehta and Dasgupta [21]	1.724855	1.724865	1.72489	NA	1.724861
Akay and Karaboga [40]	1.724852	1.741913	NA	0.031	NA
Garg [24]	1.69526388	1.69530842	1.69537060	2.836238E-5	1.69530879
Present study	1.6952471	1.6952471	1.6952471	2.192×10^{-9}	1.6952471

A variety of optimization methods have been applied to this problem by other researchers [4,5,7,10–12,14,18,20,22,23,30,34,36]. The results they obtained along with those of the proposed algorithm are given in Table 5. From the table, it has been observed that the present algorithm performs much better in comparison to other algorithms as optimal function value is better than the previous studies. The statistical result, after 30 independent runs, in terms of the best, median, mean, worst and the standard deviation obtained from the best objective value by PSO-GA approach are given in Table 6. It shows that the mean from the 30 runs performed is $f(X) = 1.6952471$ with a standard deviation of 2.192×10^{-9} . Also the worst solution found in this version is better than any of the solutions produced by any other techniques. The best solution reported by the algorithm on this version is $f(X) = 1.6952471$ corresponding to decision variable $X = [0.2057296, 3.2531200, 9.0366239, 0.2057296]$.

5.2.3. Gear train design

The gear train design problem is an unconstrained optimization. This problem has four integer variables and was introduced by Sandgren [33]. It consists of the minimization of the cost of the gear ratio of the gear train. The gear ratio is defined, for the decision variable $X = (T_d, T_b, T_a, T_f) = (x_1, x_2, x_3, x_4)$, as follows:

$$\text{Gear Train} = \frac{x_1 x_2}{x_3 x_4}$$

where T_i denotes the number of teeth of the gearwheel i and they are all integers vary in the range 12–60. The mathematical formulation is as follows

$$\text{Minimize } f(X) = \left(\frac{1}{6.931} - \frac{x_1 x_2}{x_3 x_4} \right)^2$$

$$12 \leq x_1, x_2, x_3, x_4 \leq 60 \quad ; \quad x'_i \in \mathbb{Z}^+$$

Table 7

Comparison of the best solution for gear train design problem found by different methods.

	Sandgren [33]	Kannan and Kramer [31]	Deb and Goyal [37]	Gandomi et al. [19]	Present study
$T_d(x_1)$	18	13	19	19	19
$T_b(x_2)$	22	15	16	16	16
$T_a(x_3)$	45	33	49	43	43
$T_f(x_4)$	60	41	43	49	49
Gear ratio	0.146667	0.144124	0.144281	0.144281	0.14428096
$f(X)$	5.712×10^{-6}	2.146×10^{-8}	2.701×10^{-12}	2.701×10^{-12}	2.70085×10^{-12}

Table 8

Statistical results of the gear train model (NA means not available).

Algorithms	Best	Median	Mean	Worst	Std. dev.
Sandgren [33]	5.712×10^{-6}	NA	NA	NA	NA
Kannan and Kramer [31]	2.146×10^{-8}	NA	NA	NA	NA
Deb and Goyal [37]	2.701×10^{-12}	NA	NA	NA	NA
Gandomi et al. [19]	2.701×10^{-12}	NA	1.9841×10^{-9}	2.3576×10^{-9}	3.5546×10^{-9}
Present study	2.70085×10^{-12}	9.9215×10^{-10}	1.2149×10^{-9}	3.2999×10^{-9}	8.7787×10^{-10}

Table 9

t-test: two sample assuming equal variances for Himmelblau's non-linear optimization problem.

	GA	PSO	PSO-GA
Mean	−30650.5394	−30652.8139	−30665.398373
Variance	3.663	5.803	0.058
Observations	30	30	30
Pooled variance	1.86	2.93	
Hypothesized mean difference	0	0	
Degree of freedom	58	58	
t stat	42.19	28.46	
P(T ≤ t) one-tail	0	0	
t critical one-tail	1.67	1.67	

Table 10

t-test: two sample assuming equal variances for pressure vessel design problem.

	GA	PSO	PSO-GA
Mean	6078.87552	6039.34974	5885.38205
Variance	4352.137	1179.632	0.002
Observations	30	30	30
Pooled variance	2176.070	589.817	
Hypothesized mean difference	0	0	
Degree of freedom	58	58	
t stat	16.06	24.55	
P(T ≤ t) one-tail	2.54×10^{-23}	1.23×10^{-32}	
t critical one-tail	1.67	1.67	

This unconstrained design problem has been solved previously by Kannan and Kramer [31], Sandgren [33], Deb and Goyal [37] and by Gandomi et al. [19] whose best reported results are compared with the proposed PSO-GA results in Table 7. The statistical result, after 30 independent runs, in terms of the best, median, mean, worst and the standard deviation obtained from the best objective value by PSO-GA approach are given in Table 8. It shows that the mean from the 30 runs performed is $f(X) = 1.214 \times 10^{-9}$ with a standard deviation of 8.77×10^{-10} .

5.3. Statistical analysis

In order to analyze whether the results as obtained in the above tables are statistically significantly with each other or not, we performed a t - test on pair of algorithms for the pressure vessel, welded beam design problems, etc. For this, an analysis has been conducted with the assumption that the populations have equal variances at the significance level of $\alpha = 0.05$ for the case of proposed PSO-GA results with PSO and GA results. The test has been performed against the null hypothesis that there is no difference in their population means. Under this null hypothesis, the pooled t-test has been performed in Tables 9–11 with the equality of variance and hence t-statistics values corresponding to consider the design problem are obtained. It is indicated from these tables that the values of t-stat are greater than the t-critical values. Also the p values obtained corresponding to each

Table 11

t-test: two sample assuming equal variances for welded beam design problem.

	GA	PSO	PSO-GA
Mean	1.7196679	1.7291793	1.6952471
Variance	5.48×10^{-5}	7.83×10^{-5}	4.02×10^{-18}
Observations	30	30	30
Pooled variance	2.74×10^{-5}	3.91×10^{-5}	
Hypothesized mean difference	0	0	
Degree of freedom	58	58	
t stat	18.06	20.99	
P(T ≤ t) one-tail	8.53×10^{-26}	4.41×10^{-29}	
t critical one-tail	1.67	1.67	

case is less than the significance level $\alpha = 0.05$. Thus, the two types of means differ significantly. Further, since mean of the cost functions for proposed PSO-GA approach is better than the other one and this difference is statistically significant.

6. Conclusion

This paper presents a new penalty guided hybrid approach named as PSO-GA algorithm to solve the constrained optimization problems. In the proposed approach, firstly the solution of the problem has been found by PSO and then the best solution in the current population will perform genetic algorithm operators, i.e. crossover, mutation and elitism to increase the exploration of the solution space for the next generation. To demonstrate the effectiveness and robustness of the algorithm as compared to other optimization methods, an engineering design constrained optimization problems which include pressure vessel design, welded beam design, etc., are investigated. In these optimization problems, the objective is to minimize the cost of the design subject to various non-linear constraints. The comparison results with other evolutionary algorithms demonstrate that the proposed PSO-GA algorithm proves to be extremely effective and efficient for locating the global solution or finds a near-global solution to each problem tested. Simulation results are also conducted for each problem in terms of mean, median, worst, best and standard deviation. Moreover, the standard deviations of design cost by proposed approach are pretty low, and it further implies that the approach seems reliable to solve the engineering design optimization problems. Also, the stability of the results of the proposed algorithm is justified by means of pooled t-test statistics. Thus, it is concluded that the proposed algorithm is reliable for solving the various engineering design optimization problems.

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